Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

Consider a source X that emits the following sequence of symbols:

$$\{1, 2, 6, 4, 10, 11, 10\}$$

1. Consider a quantizer Q_1, characterized by the following intervals and reconstruction levels
   - $I_1 = [-\infty, -1)$, $y_1 = -2$
   - $I_2 = [-1, 0)$, $y_2 = -0.5$
   - $I_3 = [0, +1)$, $y_3 = +0.5$
   - $I_4 = [+1, +\infty)$, $y_4 = +2$

   Plot the quantization function $y = Q_1(x)$. Encode the sequence with DPCM, quantizing the prediction residuals with Q_1 (assume that the first symbol is lossless coded). Compute the MSE distortion.
2. Design a Lloyd-Max quantizer for the source X. Initialize the algorithm with the following intervals
   - $I_1 = [-\infty, +3)$
   - $I_2 = [+3, +8)$
   - $I_3 = [+8, +10.5)$
   - $I_4 = [+10.5, +\infty)$

   Compute the MSE distortion upon convergence.

Solution

1. DPCM coding

x	1	2	6	4	10	11	10
e	-	1	3	-1	5.5	4.5	1.5
$\tilde{e}$	-	2	2	-0.5	2	2	2
$\tilde{x}$	1	3	5	4.5	6.5	8.5	10.5
$x - \tilde{x}$	0	-1	1	-0.5	3.5	2.5	-0.5

$$MSE = ((0)^2(-1)^2 + (1)^2 + (-0.5)^2 + (3.5)^2 + (2.5)^2 + (0.5)^2)/7 \quad (1)$$

2. Lloyd-Max

- Fix intervals, recompute reconstruction levels

$$y_1 = \frac{1+2}{2} = 1.5 \tag{2}$$

$$y_2 = \frac{4+6}{2} = 5 \tag{3}$$

$$y_3 = \frac{10+10}{2} = 10 \tag{4}$$

$$y_4 = \frac{11}{1} = 11 \tag{5}$$

- Fix reconstruction levels, recompute intervals

$$a_2 = \frac{5+1.5}{2} = 3.25 \tag{6}$$

$$a_3 = \frac{10+5}{2} = 7.5 \tag{7}$$

$$a_4 = \frac{11+10}{2} = 10.5 \tag{8}$$

- Fix intervals, recompute reconstruction levels

$$y_1 = \frac{1+2}{2} = 1.5 \tag{9}$$

$$y_2 = \frac{4+6}{2} = 5 \tag{10}$$

$$y_3 = \frac{10+10}{2} = 10 \tag{11}$$

$$y_4 = \frac{11}{1} = 11 \tag{12}$$

- Stop. Same reconstruction levels.

x	1	2	6	4	10	11	10
$\tilde{x}$	1.5	1.5	5	5	10	11	10
$x - \tilde{x}$	-0.5	0.5	1	-1	0	0	0

$$MSE = ((-0.5)^2 + (0.5)^2 + (1)^2 + (-1)^2 + (0)^2 + (0)^2 + (0)^2)/7 \tag{13}$$

2 Question - 5 pts

Illustrate the block diagram of a perceptual audio codec (both encoder and decoder).

Describe how the perceptual model exploits frequency masking.

Solution

See slide set `MMSP_2.2_Audio coding`.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a continuous source, and $D(R)$ the corresponding rate-distortion function, and σ_x^2 its variance.
   - [T] [F] $H(X) < \infty$
     FALSE. The entropy is undefined for continuous valued sources
   - [T] [F] $D(0) = \sigma_x^2$
     TRUE. When $R = 0$ bits are allocated, the decoder reconstructs the source as a constant signal, equal to its expected value
   - [T] [F] $R(D) \leq \frac{1}{2} \log_2 \frac{\sigma_x^2}{D}$
     TRUE. The rate-distortion function is upper bounded by the that of a Gaussian source having the same variance σ_x^2
2. Consider a speech coding system
   - [T] [F] Perceptual weighting might be used to take into account the masking effect of formants.
     TRUE. The frequency response of the perceptual weighting filter has notches corresponding to the formants of the spectrum of the underlying speech segment.
   - [T] [F] Direct quantization of prediction coefficients might lead to instability at the decoder.
     TRUE. After quantization, there is no guarantee that the shaping filter is stable, as the poles might be outside the unit circle. Therefore, it is common practice to work with reflection coefficients instead.
   - [T] [F] In open-loop vocoders, the output signal approximates the samples of the input speech waveform.
     FALSE. The output waveform can be very different from the input waveform, on a sample-by-sample basis. Yet, it sounds similar.
3. Consider a hybrid DPCM/DCT video coding system
   - [T] [F] Encoding is computationally more complex than decoding
     TRUE. This is mainly due to motion estimation.
   - [T] [F] Motion estimation is performed in the DCT domain
     FALSE. Motion estimation is typically performed in the pixel domain
   - [T] [F] Temporal redundancy is addressed using DPCM
     TRUE.
   - [T] [F] Spatial redundancy is addressed using transform coding
     TRUE.

4 Question - 12 pts

Compare the coding efficiency of DCT and KLT when compressing a ECG (electrocardiogram) signal.

- Load the signal `ecg.mat`.
- Segment the signal into non overlapping blocks of B samples ($B = 128$).
- Estimate the KLT transform matrix.
- For each block compute the DCT and the KLT, keeping only the first K samples
- Reconstruct an approximation of the input signal using only the K considered samples
- Plot the SNR for the DCT and the KLT as a function of K (use $K = 0 : 1 : 50$)
- Comment the performance of the DCT vs. KLT. Which transform scheme would you choose for an implementation on a limited-resources wireless sensor? Motivate your answer.

Solution

```
clear all
close all
clc

load ecg;

%normalize
figure()
plot(ecg);

N = length(ecg);
B = 2^7;
Nb = N/B;
range = [0:1:50];

MSE_dct = NaN*ones(length(range));
MSE_klt = NaN*ones(length(range));

% Estimate the KLT
A = [];
for ib=0:Nb-1
    %block
    xb = ecg(1+ib*B : (1+ib)*B);
    A = [A, xb];
end

R = A*A';
[V D] = eig(R);
V = fliplr(V);

ik = 1;
for k=range
    for ib=0:Nb-1
        %block
        xb = ecg(1+ib*B : (1+ib)*B);

        %dct
        xb_dct = dct(xb);

        %klt
        xb_klt = V'*xb;

        %keep only first k coeff
        xb_dct_k = [xb_dct(1:k) ; zeros(B-k,1)];
        xb_klt_k = [xb_klt(1:k) ; zeros(B-k,1)];

        %invert transform
        y_dct(1+ib*B : (1+ib)*B) = idct(xb_dct_k);
        y_klt(1+ib*B : (1+ib)*B) = V*xb_klt_k;
    end

    %compute MSE
    MSE_dct(ik) = sum((ecg' - y_dct).^2) / length(ecg);
    MSE_klt(ik) = sum((ecg' - y_klt).^2) / length(ecg);
    ik = ik+1;
end
figure();
plot(range, 10*log10(var(ecg) ./ MSE_dct));
hold on
plot(range, 10*log10(var(ecg) ./ MSE_klt),'r');
legend('DCT','KLT');
xlabel('K');
ylabel('SNR [db]');
% For the same number of K coefficients, the KLT achieves higher coding
% efficiency (larger value of SNR). The drawback is the fact that the
% matrix needs to be estimated beforehand, and transmitted to the decoder.
% This is not an issue if the signal length is much larger than the block
% size, as the number of bits / samples necessary for this operation tends
% to zero as the signal length increases. Of course, one might want to
% re-estimate the KLT when the signal characteristics change (e.g. in the
% case of anomalous ECG tracks)
```